

The collection-first ledger, বাকিবন্ধু

A Bangla-first, offline-first Android app that helps Bangladeshi micro-merchants get their credit — **বাকি** — **paid back**, not just written down. This is the single source of truth for what **v1** does.

VERSION	STATUS	BUILD	TARGET
v1 — MVP	Pre-launch · plan-ready	Solo · ১০ · ~10 weeks	Cheap Android, offline
PREPARED FOR			
Rafiq · 2026			

00 Overview

Around **8 million** Bangladeshi micro-businesses sell on credit. They track it on paper, forget who owes what, feel awkward chasing it, and quietly lose money every month. Recording isn't the pain — **getting paid back is**. Existing khata apps bury collection as a side-feature; BakiBondhu makes it the whole point.

The one job of v1

A shopkeeper can record **who owes them** money, log **payments** as they come in, always see the **true total owed**, and **nudge** a customer to pay — all in Bangla, all with no internet. Nothing else ships in v1.

Goals & success signals

GOAL	SUCCESS SIGNAL (PHASE 2)
Daily habit	Merchant records a real transaction the day <i>after</i> onboarding — unprompted
Trust in the numbers	Home total always matches reality; merchant stops cross-checking the paper khata
Less stuck money	Overdue বাকি drops over weeks of use
Frictionless onboarding	Non-technical shopkeeper is recording in under 3 minutes , zero training

Design principles. Collection is the hero · Bangla-first always · offline is the baseline, not a feature · remove, don't add · never store what you can compute · instant feel on a cheap phone.

01 Scope & the MVP fence

This is a **fence, not a wishlist**. Everything on the right is a reasonable idea and also how solo MVPs die. Say no until real users force your hand.

✓ In scope — the whole loop

- + Add a customer **FR-1**
- + Record credit — বাকি দিলাম **FR-2**
- + Record payment — টাকা পেলাম **FR-3**
- + Running balance + total owed **FR-4**
- + One-tap Bangla SMS / WhatsApp reminder **FR-5**
- + Fully offline, local-first storage **NFR-1**
- + Onboard to first বাকি < 3 min **FR-7**

✗ Out of v1 — the "not now" fence

- ✗ In-app payments / bKash–Nagad wallet
Money movement invites regulation — Phase 3+
- ✗ Inventory, barcodes, accounting, VAT
Not the collection job
- ✗ Automatic / scheduled reminders
v1 is merchant-triggered only — no gateway
- ✗ Staff logins, multi-user, web dashboard
- ✗ Any language beyond Bangla
- ✗ Editing / deleting past transactions
Append-only — correct with a reversing entry
- ✗ Forced cloud sync
v1 may ship local-only

Monetization is not in v1. The core is free forever. Paid bulk reminders, multi-shop and pro reports are a **Phase 3** build needing an SMS gateway. v1 earns nothing and costs almost nothing to run — its job is to prove daily use. Because the paid reminder is a *different* feature, Phase 0 interviews must probe willingness-to-pay directly; that assumption is load-bearing.

02 The core loop

Everything in the app serves these five steps. Build them in this order; each must feel instant — then repeat, every day.

add

01

পাটি

owe

02

বাকি দিলাম

pay

03

টাকা পেলাম

check

04

কত বাকি

nudge

05

মনে করান

Requirements are tagged **FR-n** functional · **NFR-n** non-functional · **DM-n** data rule · **AC-n** acceptance — traceable from spec → code → test.

03 The five screens

Big touch targets, Bangla-first, **one clear action per screen**. Sample figures are illustrative (নমুনা তথ্য), not a real merchant's.

SCREEN 1

Home — "money out there"

- One big number: মোট বাকি — total owed to you (৳1,24,500)
- Customers sorted by amount owed, biggest / oldest first
- Each row: name · balance · aging (৬২ দিন পার)
- Big + নতুন কাস্টমার button

SCREEN 2

Customer detail

- Running balance at top — এখন বাকি
- Full history, newest first (credit & payment)
- Three actions: বাকি দিলাম · টাকা পেলাম · মনে করান

SCREEN 3

Add transaction

- Type toggle: বাকি দিলাম / টাকা পেলাম
- Big number pad · amount is the focus
- Date defaults to আজ · optional note
- One tap সেভ করুন → back to detail

SCREEN 4

Add customer

- Name (নাম) required · phone optional but pushed
- No address, no photo, no category — on purpose
- One tap যোগ করুন

SCREEN 5

Reminder preview

- Pre-filled polite Bangla, name + amount injected
- Editable before sending
- Send via **SMS** or **WhatsApp** — merchant taps send

FIRST-RUN

Onboarding < 3 min

- No signup wall — open → add first customer
- Phone for backup asked later, not up front
- Empty state: one gentle "add your first বাকি"

Home



Customer detail



Add transaction



Reminder



OS SMS / WhatsApp

04 Data model

Two tables. That's the whole thing. **Store transactions, never store a balance — always compute it.**

CUSTOMER (PARTY)		TRANSACTION	
id	PK, generated on device (UUID)	id	PK (UUID)
name	required · duplicates allowed	customer_id	FK → customer.id
phone?	optional · needed for reminders	type	credit payment
created_at	device local time	amount	always > 0 · sign implied by type
		note? · date · created_at	note optional · date defaults today

```
# per customer — computed, never stored balance(c) = Σ credit -  
Σ payment # home headline — positive balances only total_owed =  
Σ balance(c) where balance(c) > 0
```

- DM-1 Amount stored positive; **type**, not sign, sets direction.
- DM-4 Total owed nets in **only positive** balances — a customer's advance doesn't reduce what the market owes you.
- DM-5 **Append-only**. History is never edited or deleted; correct a mistake with a reversing entry. Balances can't silently drift.
- DM-7 Aging is derived at read time from transaction dates, not stored.

UUID keys, append-only rows and timestamps make the schema **sync-friendly later** — but cloud sync is not built in v1.

05 Reminders — the one real differentiator

The feature the khata apps bury and BakiBondhu makes the point. In v1 it is **merchant-triggered**: the app opens SMS/WhatsApp with a polite, pre-filled Bangla message —

no gateway, no send cost, no spam or regulatory exposure. The merchant always presses send.

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করিম স্টোর

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আসসালামু আলাইকুম **করিম** ভাই। আপনার কাছে আমার দোকানে **৳45,000** বাকি আছে। সুবিধামতো একটু পরিশোধ করলে উপকৃত হতাম। ধন্যবাদ — **দোকানের নাম**।

SMS পাঠান

WhatsApp

Rules & edge cases

- **No phone** → action prompts the merchant to add a number first. **FR-5.5**
- **Balance ≤ 0** → reminder hidden; nothing to chase.
- **WhatsApp missing** → fall back gracefully, keep SMS.
- **Editable** → merchant may rewrite before sending.
- **No send-logging** in v1 — keeps the model at two tables.
- Tone is deliberately **polite, face-saving** — it solves the emotional awkwardness of asking, not just the admin task.

06 Non-functional requirements

As binding as the features. A correct feature that is slow, online-only, or in English fails the merchant.

ID	REQUIREMENT	BAR
NFR-1	Offline — the entire core loop works in airplane mode	Non-negotiable
NFR-2	Performance on a low-RAM phone; save feels instant	<100 ms perceived; no crashes on ৳5–8k Android
NFR-3	Usability — no training, one action per screen	Add customer + বাকি < 15 s
NFR-4	Bangla-first — whole loop needs no English	Bundle Noto Sans Bengali; verify conjuncts on real devices
NFR-5	Data safety — the ledger is their livelihood	Durable local writes; Home total = Σ balances, always
NFR-6	Privacy — data stays on device	No PII egress except the merchant-sent reminder
NFR-7	Regulatory — handles no money, moves no funds	Stays outside financial regulation in v1

07 Architecture & stack

All free-tier, chosen for one constraint above all: **scope discipline on a solo build**. v1 may ship with no backend at all.

LAYER	CHOICE	WHY
App framework	Flutter (RN if you know it better)	One codebase, smooth on cheap Android
Local storage	SQLite — drift / sqflite	Offline-first, fast on-device
Bangla type	Noto Sans Bengali , bundled	Reliable conjuncts, not device-dependent
Reminders	OS SMS composer + WhatsApp intent	No gateway, no cost, no regulation in v1
Sync / backup (later)	Supabase / Firebase free tier	Add only once merchants rely on the app

Domain logic (balance, total-owed, aging) is **pure and unit-tested** — plain functions with no UI or DB dependency, so the money math can be tested exhaustively.

08 Acceptance — definition of done

The only metric that matters in Phase 1: does a merchant open the app and record a real transaction **the next day** — **without you reminding them?** If yes, you have something. If no, no feature will save it.

v1 is "done" only when all of these hold:

- ✓ Add a customer and record a **বাকি** in under 15 seconds AC-1
- ✓ Entire core loop works fully in airplane mode AC-2
- ✓ Home total always equals the sum of every customer balance AC-3
- ✓ Reminder opens with correct name + amount, hands off cleanly AC-4
- ✓ Whole core loop completable without reading any English AC-5
- ✓ First-timer reaches first **বাকি** in under 3 minutes AC-6

- ✓

Runs without crashing on a 5–8k Android; Bangla renders right

AC-7
- ✓

No saved transaction is ever silently lost

AC-8

09 Roadmap

This document specifies **Phase 1 only**. Phase 0 must confirm the collection pain — or reshape the vertical — before the first line of app code.

WHEN	WORK	DONE WHEN
Week 4	Data + logic first. Tables, add customer/transaction, compute balance. Ugly UI is fine.	Money math provably correct; domain logic unit-tested
Week 5	Screens 1 & 2 end-to-end in Bangla on a real device	Correct balances on Home + detail
Week 6	Reminder + polish; empty states; the 3-min first-run. Feature-complete.	All five screens work; reminder pre-fills
Week 7	Test on 2–3 cheap Androids; fix crashes & Bangla. Hand to 3 friendly merchants.	Passes all acceptance criteria on real hardware
Wk 8–10	Iterate on real use. Change only what you watch merchants struggle with.	Merchants use it unprompted the next day

Your move before any code: don't open the editor first. Interview 20–30 merchants, log them in the Merchant Tracker, and let their real words confirm — or redirect — this spec. Everything after Phase 0 is execution.